

Please solve the following exercises at home prior to the tutorial:

Exercise 1

Given is a message source using the alphabet A shown in the table below with symbols $\{x_i\}$ and the respective occurrence probabilities $\{p_i\}$:

x_i	a	b	c	*	=	+
p_i	0.25	0.25	0.20	0.15	0.10	0.05

- Calculate the entropy of this data source.
- Create a binary encoding for this example. For word length, use the information content rounded up to the nearest integer. Determine the average word length and redundancy. Draw the corresponding code-tree and try to optimize the result, i.e. reduce the average word length even further.
- Determine the optimal code for alphabet A using the Huffman algorithm as well as the corresponding average word length and redundancy.
- Encode the following message using your Huffman code:

a*a+b*b=c*c

The following exercises will be done during the tutorial:

Exercise 2

Given is the alphabet $\{x_i\}$ with corresponding occurrence probabilities $\{p_i\}$ as shown in the table below:

x_i	A	E	I	O	U	Y
p_i	0.105	0.22	0.105	0.04	0.45	0.08

- Calculate the information contents $I(x_i)$ and the entropy H .
- Create the optimal binary code using the Huffman algorithm.
- What is the average word length and redundancy?
- Encode the alphabet using an optimally short code with constant word length (block code). What compression factor do we get for the Huffman code?

Exercise 3

Given is an alphabet $A = \{x_i\}$ with n symbols and occurrence probabilities p_i . What conditions must be met by n and p_i for the Huffman Code to yield a block code?